

Taipei Medical University: Supporting education excellence with a highly available digital learning environment

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About TMU

To enhance the quality of its administration and education, Taipei Medical University has adopted Google Cloud to improve the availability of academic affairs while expanding its

Established in 1960, Taipei Medical University is a leading medical education institution in Taiwan that has more than 6,000 students and six affiliated hospitals. Dedicated to offering high-quality education through continuously upgrading school software and hardware, Taipei Medical University was listed among the world's top 250 universities by the THE World University Rankings in 2022.

Industries: Education

Google Workspace
Education to build
accessible digital
environment.

Location: Taiwan

Google Cloud Premier



Google Cloud results

- Reduces system response time by 3X with the local data center of Google Cloud
- Enables high system availability by effectively distributing traffic through Cloud Load Balancing
- Supports more than 600

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About CloudMile

CloudMile is a leading AI and cloud technology company in Asia that empowers businesses to accelerate digital transformation and drive growth. Leveraging machine learning and big data analysis, CloudMile has assisted more than 500 clients with business forecasts and industrial upgrades. A Google Cloud Premier Partner, the company has become a Google Cloud Managed Service Provider (MSP) covering Taiwan, Hong Kong, and Singapore and was the Public Sector Partner of the Year for APAC Region in 2020.

Google Cloud (<https://cloud.google.com/>)

Cloud Load Balancing
(<https://cloud.google.com/load-balancing/>)

 CloudMile
Docs
(<https://workspace.google.com/products/docs/>)

ABOUT CLOUDMILE
Compute Engine
(<https://cloud.google.com/compute/>)

CloudMile is a leading AI and cloud
Google Drive
(<https://workspace.google.com/products/drive/>)

meetings
and
lectures
per day
during
COVID-19
lockdowns
with Meet

Gmail

(<https://workspace.google.com/products/gmail/>)

Google Workspace

(<https://workspace.google.com/>)

Calendar

(<https://workspace.google.com/products/calendar/>)

Sheets

(<https://workspace.google.com/products/sheets/>)

Google Meet

(<https://workspace.google.com/products/meet/>)

Education institutions in Taiwan world are exploring new method quality of teaching and learning latest technologies to campus. Digitalizing the curriculum across the board is one of such efforts, which means that institutions need highly available academic affairs systems.

Dedicated to building a world-class learning environment, Taipei Medical University (<https://eng.tmu.edu.tw/>) (TMU) has been heavily invested in upgrading its school software and hardware to improve the quality of the education provided to its more than 6,000 students. Listed among the world's top 250 universities by the THE World University Rankings (<https://www.timeshighereducation.com/world-university-rankings/taipei-medical-university>) in 2022, the university started deploying its websites, administration systems, databases and online learning platforms in a hybrid cloud environment several years ago, and has been using Google Workspace for Education (https://edu.google.com/intl/en_ALL/workspace-for-education/editions/overview/)

as the backbone of its school email and office collaboration systems.

TMU previously incorporated public cloud platforms offered by Taiwanese and foreign providers in its hybrid cloud architecture to run its admission, course registration and human resource systems. Since the foreign public cloud provider didn't have a data center in Taiwan, the integration between the public cloud and the university's on-prem IT structure was not ideal, which led to poor connectivity and stability. Moreover, the Taiwanese cloud provider couldn't offer sufficient scalability and system integration. During the traffic peak periods like the beginning of a semester, TMU's academic affairs system often experienced downtime for hours.

"The public cloud platform that we previously used could not handle thousands of access requests to our systems at the same time, so our administration systems frequently crashed when students tried to register their courses or apply for our programs," explains Kevin Chang, IT director at TMU. "But these are exactly the times when we need the systems to function well the most, because students have to complete these administration procedures within a tight deadline."

To improve user experience, TMU continued in its search for a public cloud platform that could accommodate its needs. After comparing different offerings, the university landed on Google Cloud in late 2021. With a data center in Taiwan, Google Cloud could support better integration with its local

on-prem IT environment, resulting in faster data transfers. Along with the migration, TMU also wanted to increase the quality of remote teaching during COVID-19 lockdowns by expanding its use of Google Workspace for Education.

"We've done a comparison test, and found that Google Cloud not only has a data center in Taiwan, but also stood out by supporting the best performance and user experience," says Chang. "At the same time, we've had excellent experience from using Google Workspace for Education for more than 10 years and wanted to leverage its latest technologies to create a better digital learning environment."

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A well-integrated hybrid cloud environment supporting quick system responses

As one of the first higher education institutions in Taiwan that adopts both Google Cloud and Google Workspace for Education, as well as an organization that has two campuses and six affiliated hospitals,

TMU required extensive external technical assistance to map out the migration and operation plan. Working with Google Cloud partner agency CloudMile (<https://mile.cloud/>), who is experienced in cloud migration, TMU was able to design a new hybrid cloud architecture that seamlessly integrates on-prem and Google Cloud resources, and set up subnets for better management on Google Workspace for Education.

To accelerate the adoption process, CloudMile also held four training sessions that walked the TMU team through the Google Cloud ecosystem. As a result, TMU successfully completed migrating all its administration systems to Google Cloud and upgraded its Google Workspace for Education plan to Google Workspace for Education Plus in two months.

"CloudMile shared with us the best practices for system integration and restructuring. During the migration process, the CloudMile team guided us through each step and shared very useful technical tips," says Chang.

The first significant difference that the TMU noticed after the migration was the connection speed. It used to take the human resources system 20-30 microseconds to respond and more than 20 seconds to load an image. After the migration to Google Cloud, the system always responds within 10 microseconds and only needs three to five seconds for image loading.

Eliminating system delays and crashes with Compute Engine and Cloud Load Balancing

The student admission and course registration periods are the times when universities' administration systems encounter higher risk of errors and downtime. During these periods, schools with insufficient computing resources might see their systems having serious response delays or being fully down for five to six days. When a system is down, it requires a lot of time and effort to clear the issue.

Now, TMU uses the virtual machines (VMs) on Compute Engine (<https://cloud.google.com/compute>) to run the two systems and leverages Cloud Load Balancing (<https://cloud.google.com/load-balancing>) to distribute traffic among VMs. To establish the most cost-efficient and error-free practice, the university also set up a standard operating procedure for adjusting the number of VMs according to needs, which went through a number of pressure tests before the systems were launched on Google Cloud. Now, the VMs on Compute Engine can be instantly scaled up by more than 15X. Cloud Load Balancing is also able to redirect incoming traffic in time before a VM is overloaded. TMU's admission and course registration systems no longer experience any crashes or delays since the migration.

"The load balancing mechanism of Compute Engine and Cloud Load Balancing is very effective, because it can quickly launch a new VM," says Rock Hsu, IT deputy director at TMU. "Since we migrated our

systems to Google Cloud, no issues have occurred during the periods when a huge number of computing resources are needed, such as student admission and course registration."

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Supporting university administration and enabling online classes during COVID-19 and beyond

TMU has been using Gmail (<https://workspace.google.com/products/gmail/>) as its school email system, Drive (<https://www.google.com/drive/>) to store all its internal files, and the single sign-on feature of Google Workspace for Education across all its school systems for easier account access for more than 10 years. It has also been relying on the concurrent editing feature of Docs (<https://workspace.google.com/products/docs/>) and Sheets (<https://workspace.google.com/products/sheets/>) to realize efficient office collaboration. When the COVID-19 lockdown measures made remote working obligatory in Taiwan in 2021, the team was able to quickly adapt to the new workplace model without sacrificing any productivity.

However, the university was able to enhance its Google Workspace for Education use cases after upgrading to Google Workspace for Education Plus. Today, TMU is able to hold video meetings with up to 500 participants on Meet (<https://meet.google.com/>), which has helped maintain its teaching quality even in a remote setting. By providing comprehensive technical support and training, the university successfully assisted its teachers and students to get the hang of the new learning platform and fully enjoy the extra benefits brought by the advanced online meeting features like breakout rooms and polling. During the pandemic lockdown periods, more than 600 lectures, group discussions and meetings at TMU took place on Meet every day.

On top of that, the high integration among Meet, Gmail and Calendar (<https://workspace.google.com/products/calendar/>) means that TMU's teachers, students and staff can easily schedule and update meetings without too much back-and-forth communication. The feature is particularly helpful for the university, since it has many lectures taken by hundreds of students.

"Google Workspace for Education has been supporting our administration and academic workloads for more than a decade, and the advantages of using it became even more obvious during COVID-19, because our teachers, students and staff can easily carry on their daily work as usual no matter where they are," says Marian Wan, IT deputy director at TMU.

Having benefited from the extensive support of Google Workspace for Education Plus for online classes, TMU has made remote teaching a norm for all its lectures to increase the flexibility of class arrangements across different campuses. It has also created more interactive class experiences through adopting Chromebooks

(<https://www.google.com/chromebook/>) to build smart classrooms. Since Chromebook is fully integrated with Google Workspace for Education, TMU's teachers and students can easily have discussions and co-edit files on the tablets during classes.

Building a cloud-based, highly accessible learning environment

TMU continues to build a digital learning environment that enables its teachers to teach creatively and its students to learn autonomously by introducing more new technologies. The university has merged all its academic affairs systems on Google Cloud into Education Cloud, on which it will launch a personalized student learning portal embedded with Calendar, so that its students can track their course schedules more easily.

On the management side, TMU plans to leverage Looker (<https://cloud.google.com/looker>) in the near future to get a better picture of the university's operation, while using Dialogflow (<https://cloud.google.com/dialogflow>) to build chatbots that can provide automated support. It will also adopt Google Cloud Armor (<https://cloud.google.com/armor>) to strengthen the security of its academic affairs systems. To

facilitate its clinical research, the university will use BigQuery (<https://cloud.google.com/bigquery>) and Dataflow (<https://cloud.google.com/dataflow>) to accelerate data processing.

"Our goal is to establish a high-quality education environment that brings all the benefits of digitalization to our teachers and students," says Wan. "With Google Cloud, Chromebooks and Google Workspace for Education Plus, we have cleared the technical obstacles of our school systems and successfully implemented hybrid classes. Moving forward, we'll leverage more advanced technologies of Google Cloud to continue enhancing the teaching and learning quality of our university."

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